Balloon Inflating Deployment and Disassembly.

Equipment Check: Make sure all equipment, manpower safety aspects are organized before starting the whole process. Inflate the balloon using the nozzle provided by the Helium tank manufacturer. If the tank is under heavy pressurization use a pressure reduction valve.

Inflation: Have two people during the inflation. One should hold the balloon from trying to buoy and other to hold the balloon to the nozzle closely watch for leaks of Helium at the valve stems. Using earplugs while inflating is highly recommended. The template should be subsequently inserted to get the desired diameter of 3ft. A third person would be ideal for this process.

Make sure to prevent sharp edges of the tank or on the ground from impinging on the balloon surface.

Clamping and Securing Balloon: Once balloon is inflated to the desired diameter fold the stem of the balloon. (Use width of you three middle fingers). Clamp the folded edge. Now using masking tape wrap the folded area around the stem.

Tying Spectra to Balloon: Take the Spectra 2000 and wrap it around the stem of the balloon. Try at best to tie around the taped area. This prevents the spectra from cutting the balloon surface. Wind the spectra around the stem many times before putting the fisherman's knot. Again wrap the knotted area and the knot with masking tape. This would further prevent the spectra from slipping in case of a knotting failure.

Use the capture net to place the balloon if sufficient manpower is not available. Until the Skyscout and telescope tripod assembly is ready to use.

Deployment: Deploy the balloon gently while loosely holding the tether line. If wind gusts deploying the balloon would be chaotic and difficult. Deploying the balloon faster to about 100 feet – 200 AGL we would observe a reduction in chaotic movement of the balloon. Gently slow down balloon deployment when required altitude is reached. This is to prevent sudden jolt of the tether line as you know the tether could not take over 20lbs of force.

During the whole deployment procedures watch for cycle computer readings.

Recovery: Recovering the balloon should be done gently without jolting the tether line. Maintain a steady speed while winding back the tether. One person should crack the winch other guides the line by loosely holding the line. Now the balloon is captured use the net to place the balloon if a retest is to take place.

Deflating Balloon: Use a Swiss knife or small scissors to cut the masking tape trough the folded area of the stem. Eventually you would cut through the tether knots. Make sure you don’t pierce the balloon wall. Now the deflated balloon could be reused.

Remember: Safety doesn't happen by accident. "Safety First" is "Safety Always." Prepare and prevent, don't repair and repent. ~Author Unknown.

Faisal Buharie. ***March 14, 2008.***